CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

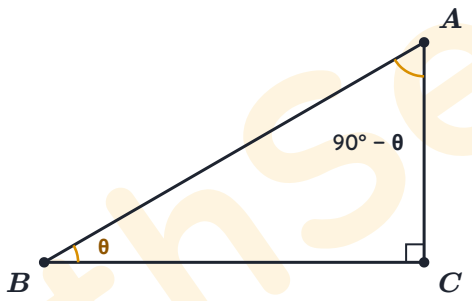
Chapter 25: Complementary Angles

Topic-wise short notes with solved examples

The trigonometrical ratios of complementary angles and their use in simplifying expressions, proving results and solving equations without tables.

1 Complementary angles

- Two angles are **complementary** if their sum is 90°. The complement of θ is $90^\circ - \theta$.
- In a right-angled triangle the two acute angles are complementary. The perpendicular for one is the base for the other.



The acute angles at B and A are complementary

- This gives the **ratios of complementary angles**:

Ratio of $90^\circ - \theta$	Equals
$\sin(90^\circ - \theta)$	$\cos\theta$
$\cos(90^\circ - \theta)$	$\sin\theta$
$\tan(90^\circ - \theta)$	$\cot\theta$
$\cot(90^\circ - \theta)$	$\tan\theta$
$\sec(90^\circ - \theta)$	$\operatorname{cosec}\theta$
$\operatorname{cosec}(90^\circ - \theta)$	$\sec\theta$

- In words: each ratio changes to its **co-ratio**: sine with cosine, tangent with cotangent, secant with cosecant.
- So if $A + B = 90^\circ$, then $\sin A = \cos B$, $\tan A = \cot B$ and $\sec A = \operatorname{cosec} B$.

2 Evaluating without tables

- Look for pairs of angles that add up to 90°. Change one of each pair to its complement.

Example 2.1 Evaluate $\frac{\sin 35^\circ}{\cos 55^\circ}$.

SOLUTION

$\cos 55^\circ = \cos(90^\circ - 35^\circ) = \sin 35^\circ$

1

Example 2.2 Evaluate $\tan 27^\circ - \cot 63^\circ$.

SOLUTION

$\cot 63^\circ = \tan(90^\circ - 63^\circ) = \tan 27^\circ$

0

Example 2.3 Evaluate $\frac{\cos 70^\circ}{\sin 20^\circ} + \frac{\cos 59^\circ}{\sin 31^\circ} - 8\sin^2 30^\circ$.

SOLUTION

$1 + 1 - 8 \times \frac{1}{4}$

0

Example 2.4 Evaluate $\sin^2 25^\circ + \sin^2 65^\circ$.

SOLUTION

$\sin 65^\circ = \cos 25^\circ$, so the sum is $\sin^2 25^\circ + \cos^2 25^\circ$

1

Example 2.5 Evaluate $\tan 10^\circ \tan 20^\circ \tan 45^\circ \tan 70^\circ \tan 80^\circ$.

SOLUTION

$\tan 80^\circ = \cot 10^\circ$ and $\tan 70^\circ = \cot 20^\circ$. Each pair multiplies to 1.

$1 \times 1 \times \tan 45^\circ = 1$

Example 2.6 Evaluate $\frac{\sec 42^\circ}{\operatorname{cosec} 48^\circ} + \frac{2\tan 38^\circ}{\cot 52^\circ}$.

SOLUTION

$1 + 2$

3

3 Expressing in terms of angles below 45 degrees

- Any ratio of an angle between 45° and 90° can be written as a ratio of an angle between 0° and 45°.

Example 3.1 Express $\sin 67^\circ + \cos 75^\circ$ in terms of angles between 0° and 45°.

SOLUTION

$\cos 23^\circ + \sin 15^\circ$

Example 3.2 Express $\tan 81^\circ + \sec 64^\circ$ in terms of angles between 0° and 45°.

SOLUTION

$\cot 9^\circ + \operatorname{cosec} 26^\circ$

4 Solving equations

- If $\sin A = \cos B$ with A and B acute, then $A + B = 90^\circ$. The same holds for $\tan A = \cot B$ and $\sec A = \operatorname{cosec} B$.

Example 4.1 If $\sin 3\theta = \cos(\theta - 6^\circ)$, where 3θ and $\theta - 6^\circ$ are acute, find θ.

SOLUTION

$3\theta + \theta - 6^\circ = 90^\circ$, so $4\theta = 96^\circ$

$\theta = 24^\circ$

Example 4.2 If $\tan 2A = \cot(A - 18^\circ)$, where $2A$ is acute, find A.

SOLUTION

$2A + A - 18^\circ = 90^\circ$

$A = 36^\circ$

Example 4.3 If $\sec 4A = \operatorname{cosec}(A - 20^\circ)$, where $4A$ is acute, find A.

SOLUTION

$5A - 20^\circ = 90^\circ$

$A = 22^\circ$

5 Proving results

- Replace one angle by 90° minus the other, or use the angle sum of a triangle.
- In triangle ABC, $A + B + C = 180^\circ$, so $\dfrac{B + C}{2} = 90^\circ - \dfrac{A}{2}$.

Example 5.1 If A, B and C are the angles of a triangle, prove that $\sin\frac{B+C}{2} = \cos\frac{A}{2}$.

SOLUTION

$\frac{B+C}{2} = 90^\circ - \frac{A}{2}$

$\sin\left(90^\circ - \frac{A}{2}\right) = \cos\frac{A}{2}$

Example 5.2 Prove that $\frac{\cos(90^\circ-\theta)\sin(90^\circ-\theta)}{\tan(90^\circ-\theta)} = \sin^2\theta$.

SOLUTION

LHS $= \frac{\sin\theta\cos\theta}{\cot\theta} = \sin\theta\cos\theta \times \frac{\sin\theta}{\cos\theta}$

$\sin^2\theta$

Example 5.3 Prove that $\sin 48^\circ \sec 42^\circ + \cos 48^\circ \operatorname{cosec} 42^\circ = 2$.

SOLUTION

$\sec 42^\circ = \operatorname{cosec} 48^\circ$ and $\operatorname{cosec} 42^\circ = \sec 48^\circ$

$\sin 48^\circ \times \frac{1}{\sin 48^\circ} + \cos 48^\circ \times \frac{1}{\cos 48^\circ} = 2$

Example 5.4 If $A + B = 90^\circ$, prove that $\sqrt{\frac{\tan A \tan B + \tan A \cot B}{\sin A \sec B}} = \sec A$.

SOLUTION

$\tan B = \cot A$, $\cot B = \tan A$, $\sec B = \operatorname{cosec} A$. Numerator $= 1 + \tan^2 A = \sec^2 A$. Denominator $= \sin A \operatorname{cosec} A = 1$.

$\sqrt{\sec^2 A} = \sec A$

6 Common mistakes

- Changing the angle but keeping the same ratio: $\sin 70^\circ$ is $\cos 20^\circ$, not $\sin 20^\circ$.
- Pairing secant with cosine. The co-ratio of secant is cosecant.
- Using the rule for angles that do not add up to 90°.
- Changing **both** angles of a pair, which brings the expression back to where it started.
- In equations, writing $A = B$ from $\sin A = \cos B$. The correct step is $A + B = 90^\circ$.